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AI for Bharat Hackathon

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Team Name: Sudharshan AI

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Problem Statement: Combating the "Digital Arrest" epidemic and real-time UPI coercion using privacy-first, agentic financial intelligence.

Brief about the Idea:

The Crisis: India lost ₹22,495 crores to cyber fraud in 2025 alone - a 24% surge in cases, with 7.4 lakh complaints filed to NCRP. Digital Arrest scams alone drained over ₹3,000 crore, with 1 in 5 UPI families affected. And those are just the reported numbers. <u>85% of victims fail to act within the critical "Golden Hour"—the window before funds are withdrawn or laundered.</u>

Sudarshan-AI is a real-time, on-device financial guardian that detects psychological coercion and fraud patterns during live UPI transactions. By combining Behavioral Biometrics with Agentic RAG, it prevents funds from leaving the victim's account during the "Golden Hour" of a crime, even if the user is being manipulated or isolated by scammers.

- **Differentiation:** Current systems are reactive (reporting after loss). Sudarshan is proactive (intervention before the COMMIT). Unlike bank filters that only see transaction amounts, we analyze the human state (behavioral signals, typing patterns, tremors, hesitation and voice detection[phase 2]).

- **How it solves the problem:** It implements a "Silent Circuit Breaker"—holding funds in a secure line without alerting the scammer, giving law enforcement a required and reasonable tactical window to intervene.

- **USP:** “Zero-Knowledge” Safety. We analyze coercion signatures locally on the edge (no audio uploaded to cloud) to satisfy the **DPDP Act 2023**.

The Why, How and What:

- **Why AI is required?** → Rule-based systems only see amounts; AI reasons through behavioral psychology
- **How AWS services are used?** → Bedrock for AI reasoning, Lambda for compute, Step Functions for circuit breaker, DynamoDB for baselines
- **What value AI adds to UX?** → Invisible protection, <1s intervention, explainable decisions

List of features offered by the solution

Core Features (Will demo):

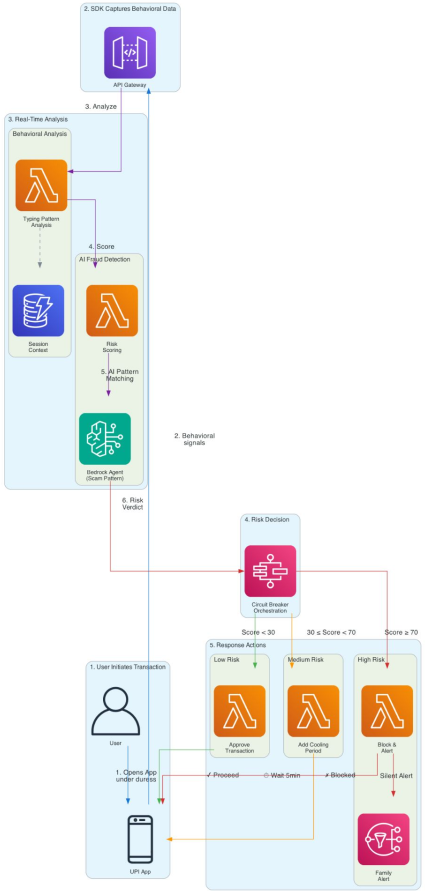
1. **Agentic Fraud-Script Analysis** - Bedrock Agents analyze transaction context against known scam patterns
2. **Silent Circuit Breaker** - Step Functions orchestrate fund hold without alerting scammer
3. **Behavioral Biometrics** - Typing patterns, navigation flow anomaly detection
4. **Emergency Duress PIN** - Secret PIN triggers silent alert to authorities

Roadmap Features (Future Vision):

1. Voice Sentiment Analysis (Phase 2 - Mobile SDK)
2. On-Device Edge AI for offline protection (Phase 3)
3. Nitro Enclaves for enterprise deployment (Phase 3)

Process flow diagram:

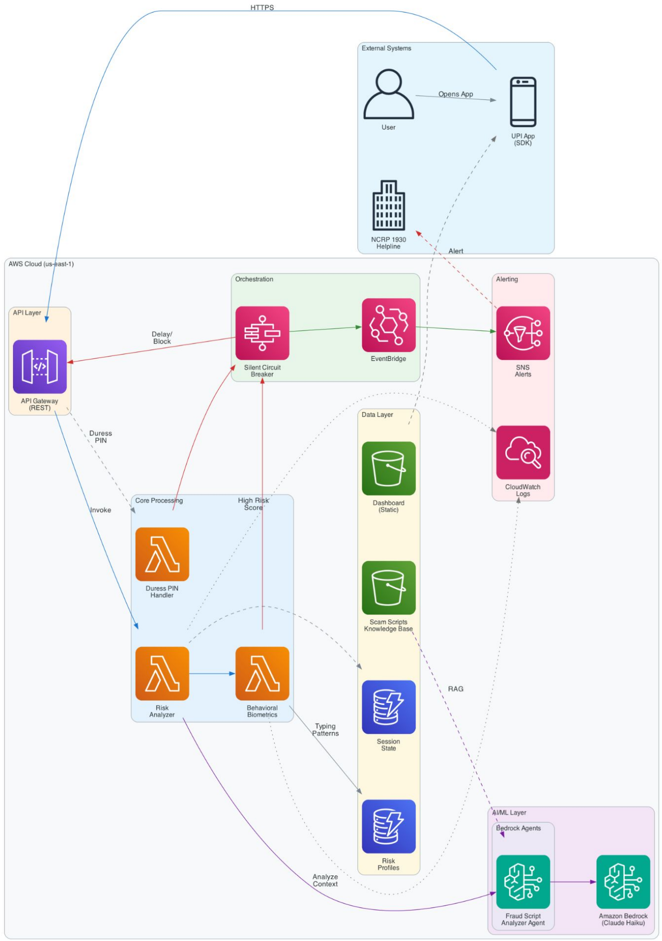
1. User opens UPI app while on a scam call
2. SDK detects abnormal typing patterns (hesitation, tremors)
3. Bedrock Agent analyzes transaction context against scam script database
4. Risk score triggers Silent Circuit Breaker (Step Functions)
5. Transaction held + family/authorities silently alerted
6. Dashboard shows real-time risk visualization



Sudharshan-AI: Fraud Detection Flow

Architecture diagram

of the proposed solution:



Sudharshan-AI: Real-Time UPI Fraud Prevention

Technologies & Implementation Costs in the solution:

Service	Purpose	Free Tier Charges
AWS Bedrock Agents (Nova)	Fast, cheap classification	Pay-per-use ($0.25/1M tokens)
AWS Bedrock Agents (Haiku)	Complex agentic reasoning	Minimal usage for demos
AWS Lambda	Serverless compute	1M requests/month Free
Amazon DynamoDB	Transaction metadata, baselines	25GB Free
AWS Step Functions	Workflow orchestration	4K transitions/month Free
Amazon API Gateway	REST API endpoints	1M calls/month Free
Amazon S3	Static web hosting, logs	5GB Free
Amazon SNS	Alert notifications	1M publishes Free

Estimated Prototype Cost: $15 - 30 (with Free Tier benefits)

Technical Differentiators:

Current Bank Systems	Sudarshan-AI
Detection: Rule-based: "Amount > X = Block"	**Agentic AI: Reasons through scammer psychology**
Reactive: Report after money lost	**Proactive: Intervene before transaction commits**
10-20 mins (manual helpline)	**<1 second (automated intervention)**
Invasive: Call recording, screen monitoring. Retrofitted for DPDP	**Privacy-first: Behavioral signals only, no raw data stored. Built for DPDP Act 2023 from day one**
Static rules, easily bypassed	**Dynamic: Learns new scam patterns via Bedrock**
English-centric interfaces	**Designed for 250M+ non-English users**

Business Impact & Feasibility:

- **Target Customers:** Banks, Fintechs (PhonePe, Google Pay, Paytm)
- **Business Model:** Safety-as-a-Service API - per-transaction micro-fee
- **Alignment:** IndiaAI Mission, RBI Advisory on real-time fraud prevention
- **Sustainability:** Reduces bank fraud losses → clear ROI for customers

Impact Metrics:

- **250M+** potential beneficiaries (non-English speaking UPI users)
- **Reduce MTTC** (Mean Time To Contain) from 10+ mins to <1 second
- **Potential:** To prevent ₹500+ crore annually based on current fraud volumes
- **Systemic impact:** Restore trust in Digital Public Infrastructure

Roadmap:

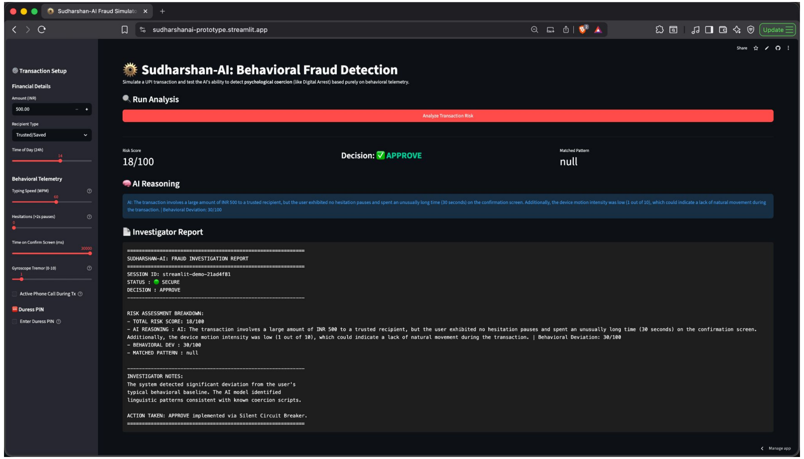
Phase	Timeline	Deliverable
Prototype	March 2026	Core MVP with Bedrock + Step Functions
Phase 2	Q2 2026	Mobile SDK with voice sentiment
Phase 3	Q2 2026	Edge AI for offline, Nitro for enterprise
Pilot	Q3 2026	B2B pilot with regional banks

Risk Mitigation:

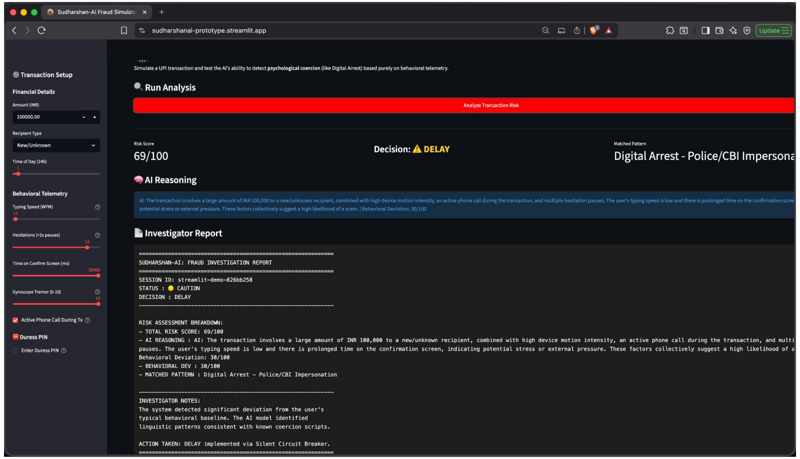
- **User can override with biometric + confirmation**
- **Trusted contacts whitelist for recurring payments**
- **Continuous learning reduces false positives over time**

Snapshots of the prototype:

Note: This is a **tunable prototype simulator** for validating detection logic. In production, behavioral signals (typing patterns, device tremor, call status) would be captured in **real-time from device sensors** via a Mobile SDK **integrated with UPI interfaces**. This interface allows **manual parameter adjustment** to demonstrate the system's response across different fraud scenarios.

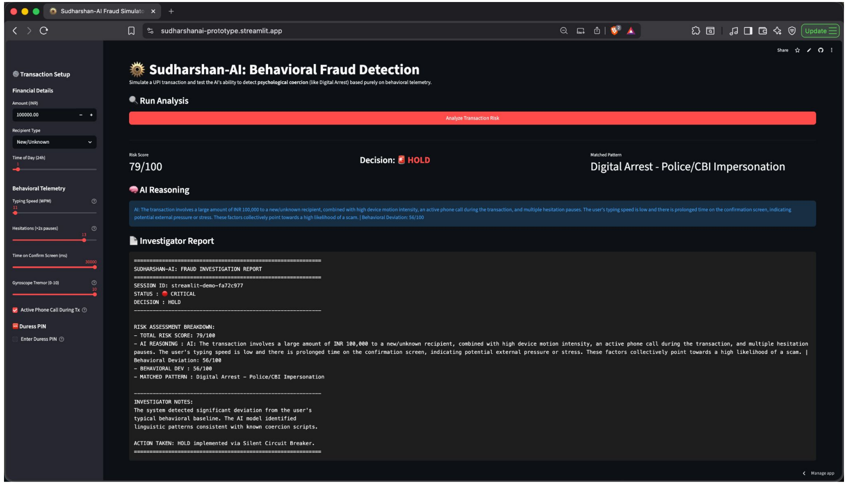


Scenario 1: Low Risk

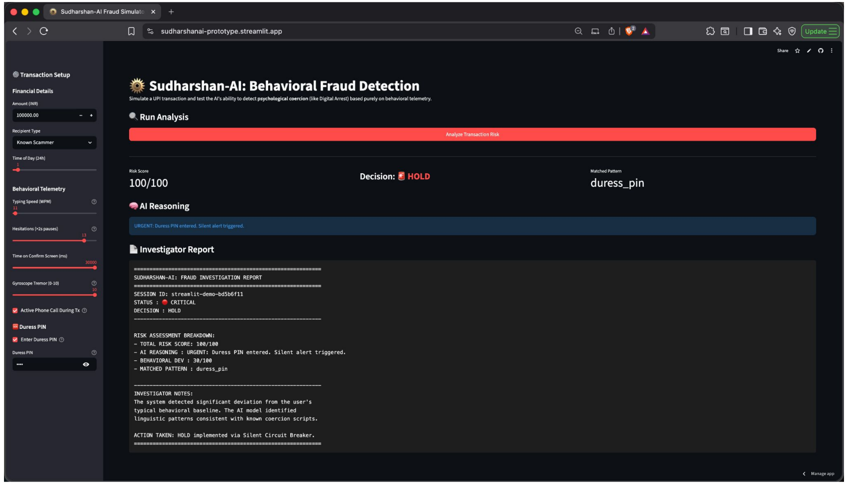


Scenario 2: Medium Risk

Snapshots of the prototype:



Scenario 3: High Risk



Scenario 4: Very High Risk (Duress Triggered)

Duress PIN: When a user enters the secret Duress PIN (e.g., 9999) under coercion, the system forces a risk score of 100/100 and silently alerts trusted contacts — while the transaction appears to proceed normally to the scammer.

Prototype Performance report/Benchmarking:

Metric	**Target**	**Achieved**
Risk Scoring Latency	< 500ms	✅ ~400-600ms (Bedrock inference)
End-to-End API Latency	< 1 second	✅ < 1s for most requests
Detection: Digital Arrest	High confidence	✅ Correctly identifies coercion signals
Detection: Known Scammer	Immediate HOLD	✅ Risk score 90+
False Positive (normal txn)	< 5% flagged	✅ Normal transactions score < 30
Duress PIN	100% trigger rate	✅ Verified in mock tests
Behavioral Deviation	Detects anomalies	✅ Score increases with signal deviation

Additional Details/Future Development (if any):

Expansion Use Cases:

- Elder fraud protection
- OCEN loan shark coercion detection
- Investment scam/Honeytrap detection
- Mule account identification for banks

Business Model: Safety-as-a-Service API — per-transaction micro-fee for banks & fintechs

Test Scenarios Verified:

1. Safe transaction → APPROVE (score ~15-25)
2. Coercion pattern → HOLD (score ~85-95)
3. Known flagged recipient → HOLD (score ~90+)
4. Duress PIN → Instant silent alert (score = 100)

Prototype Assets:

- ➔ **GitHub Public Repository Link:** https://github.com/charanpool/sudharshan-ai/
- ➔ **Demo Video Link:** https://drive.google.com/file/d/1omNeGjdtG7T0ZjlTo460p3RwVBUBgB-j/view?usp=sharing
- ➔ **Live Prototype:** https://sudarshanai-prototype.streamlit.app/

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